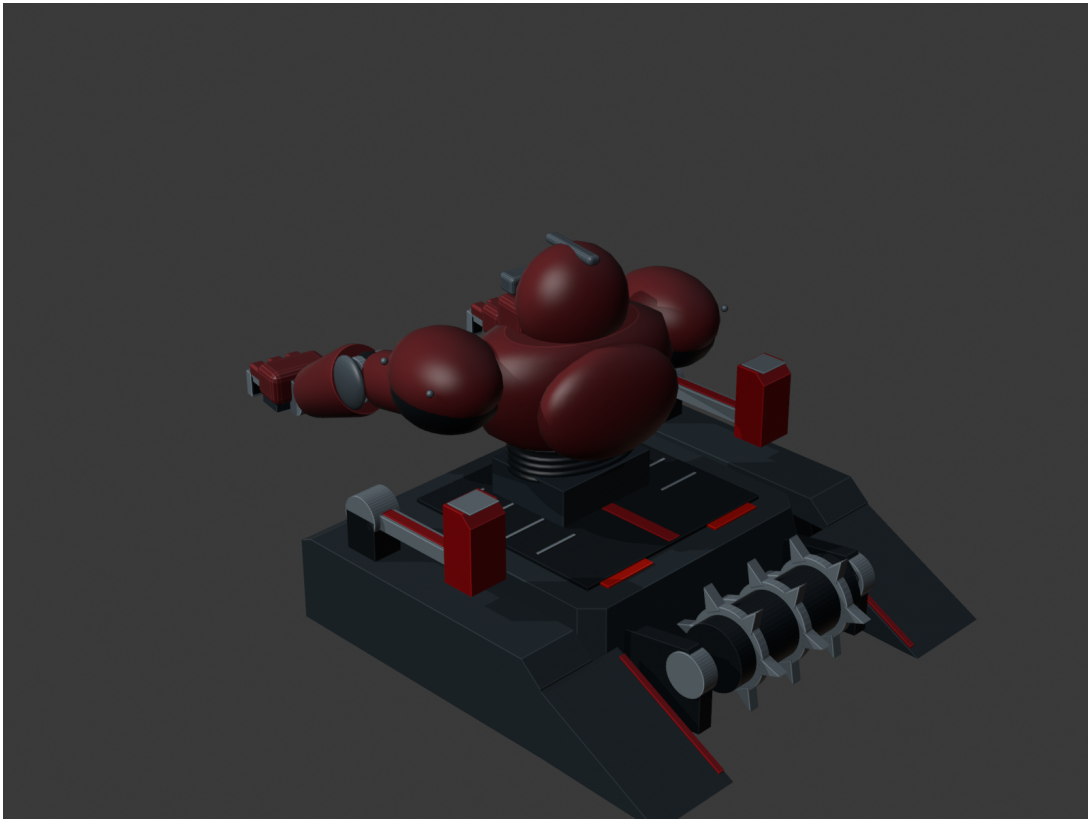


SENTINELT

Robo Wars RC Combat Robot

Holistic Build Document

Designed and developed by Thato Glen Assegai



Verified CAD envelope: 318.16 mm x 495.00 mm x 255.41 mm | Footprint audit: 180/180 frames PASS

1. Project Ownership and Overview

SentinelT is my independently designed Robo Wars RC combat robot. I developed the mechanical concept, Blender CAD model, dimensional-verification scripts, 2.4 GHz control framework, power and safety architecture, component plan and technical documentation represented in this repository.

The current build is complete as a digital engineering design. Physical fabrication and hardware validation remain separate implementation stages, so measured mass, current draw, thermal performance and physical E-Stop response are not claimed as completed results.

Design item	Current verified state
CAD width (X)	318.16 mm
CAD length (Y)	495.00 mm
CAD height (Z)	255.41 mm
Footprint verification	180 / 180 frames PASS
Engineering mass budget	4.650 kg
Design margin below 5 kg	0.350 kg
Control architecture	Human-operated 2.4 GHz RC

2. Robo Wars Design Requirements

SentinelT is engineered around the Robo Wars operating envelope and safety requirements: a maximum 500 mm x 500 mm footprint, mass below 5 kg, 2.4 GHz human-operated RC, accessible main ON/OFF isolation, an Emergency Stop power-disable mechanism and no projectiles, flames or liquids.

3. Mechanical Design

SentinelT uses a compact armored lower chassis with protected drive areas, sloped external protection, a central service volume, removable access panels and an articulated upper structure.

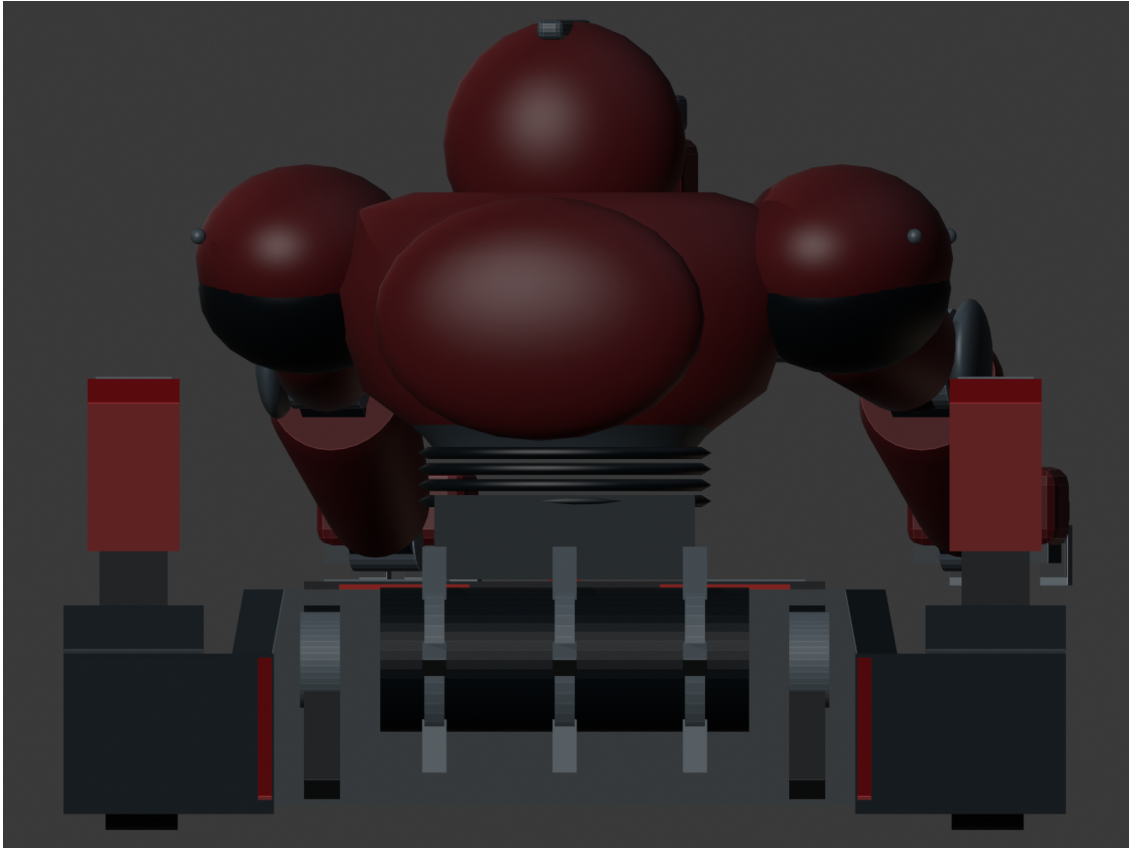


Figure 1. Front CAD view of SentinelT.

3.1 Fabrication and materials

The current material plan uses 6 mm HDPE sheet for the chassis and shared armor panels, with metal hardware used where load transfer, motor housings, shafts, brackets, electrical terminals and fasteners require it. Bolted construction keeps the design serviceable and compatible with the 4.650 kg engineering mass budget.

4. Electronic Design

The electronic system is organized around a 3S battery, high-current power distribution, 2.4 GHz RC control and independent safety isolation. The component plan is treated as a matched engineering system and remains subject to final physical validation.

Subsystem	Selected / planned component
Drive motors	2 x Pololu 4743 50:1 12 V 200 RPM 37D metal gearmotors
Drive wheels	125 mm all-terrain pair with 6 mm shaft fittings
Drive controller	HobbyWing QuicRun WP 880 dual brushed ESC
RC link	FlySky FS-i6X transmitter + X6B receiver, 2.4 GHz
Battery	BEAT 5000 mAh 11.1 V 3S LiPo, XT60
Power distribution	100 A 4-way terminal block
Main isolation	12 V 100 A battery isolator candidate
E-Stop control	Emergency pushbutton + separate high-current relay / contactor candidate
Power wiring	10 mm ² silicone red / black high-current wiring set

5. Power and E-Stop Safety Architecture

The main ON/OFF function and Emergency Stop function are explicit parts of the power path. The emergency pushbutton is not assumed to interrupt full traction or auxiliary current directly; it commands a separate high-current power-isolation device.

SentinelT Power & E-Stop Architecture

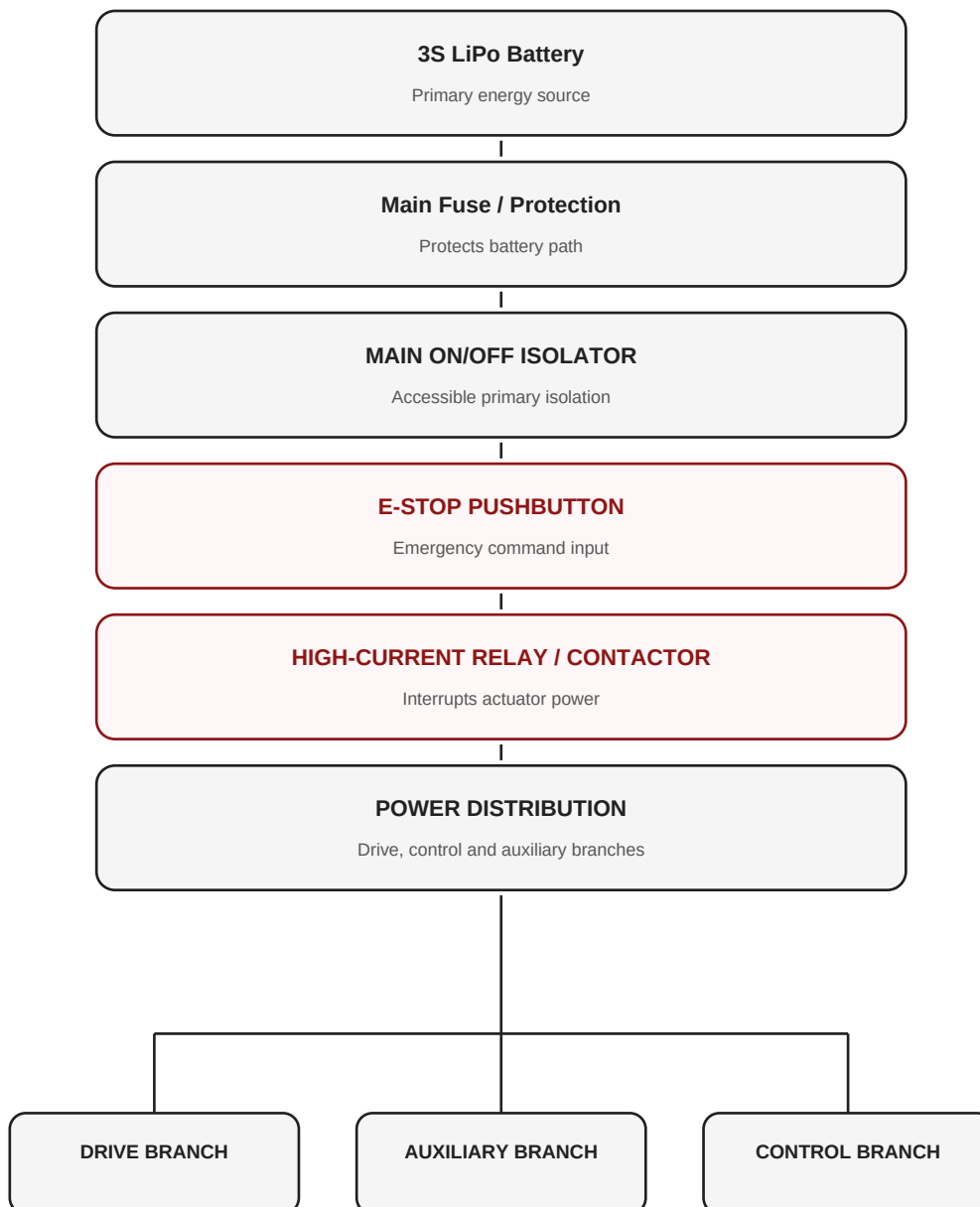


Figure 2. SentinelT power, control and E-Stop architecture.

6. 2.4 GHz RC Control

SentinelT is human-operated through a 2.4 GHz transmitter and receiver. The operator provides throttle, steering and auxiliary-enable commands. Throttle and steering are converted into differential-drive commands, while auxiliary actuation remains disabled by default.

7. Programming and Framework Design

The software architecture is written specifically for Robo Wars operation. Human RC input is always gated by hardware safety state, signal validity and the auxiliary-enable command.

SentinelT 2.4 GHz RC Control Flow

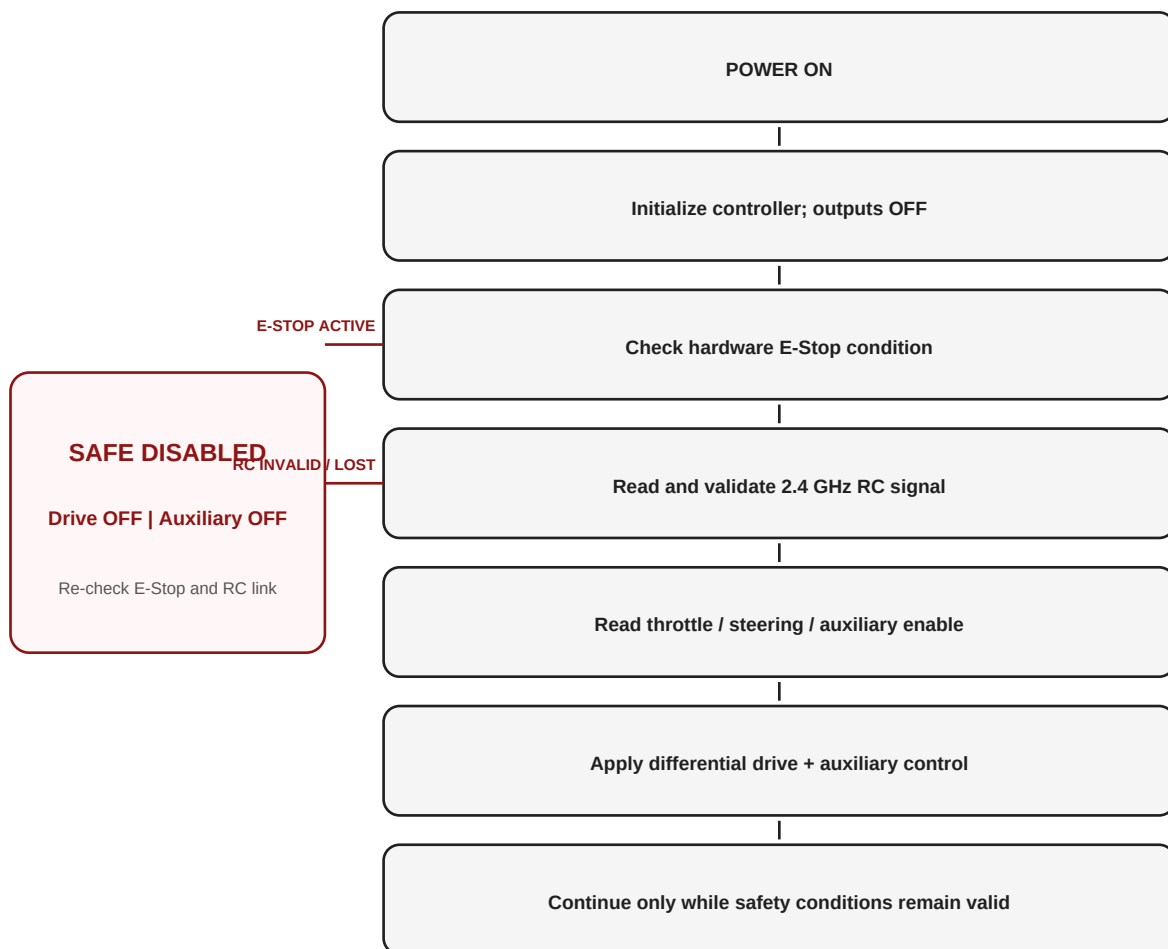


Figure 3. SentinelT safe-state control-flow logic.

8. Simulation and Digital Verification

I used Blender and Python-based auditing to verify SentinelT before fabrication. The footprint script evaluates visible mesh world-space bounds across frames 1-180 and compares the X and Y dimensions with the 500 mm x 500 mm Robo Wars footprint.

Audit parameter	Verified result
Maximum X	318.16 mm
Maximum Y	495.00 mm
Maximum Z	255.41 mm
Frames checked	180
Failed frames	0
Result	PASS

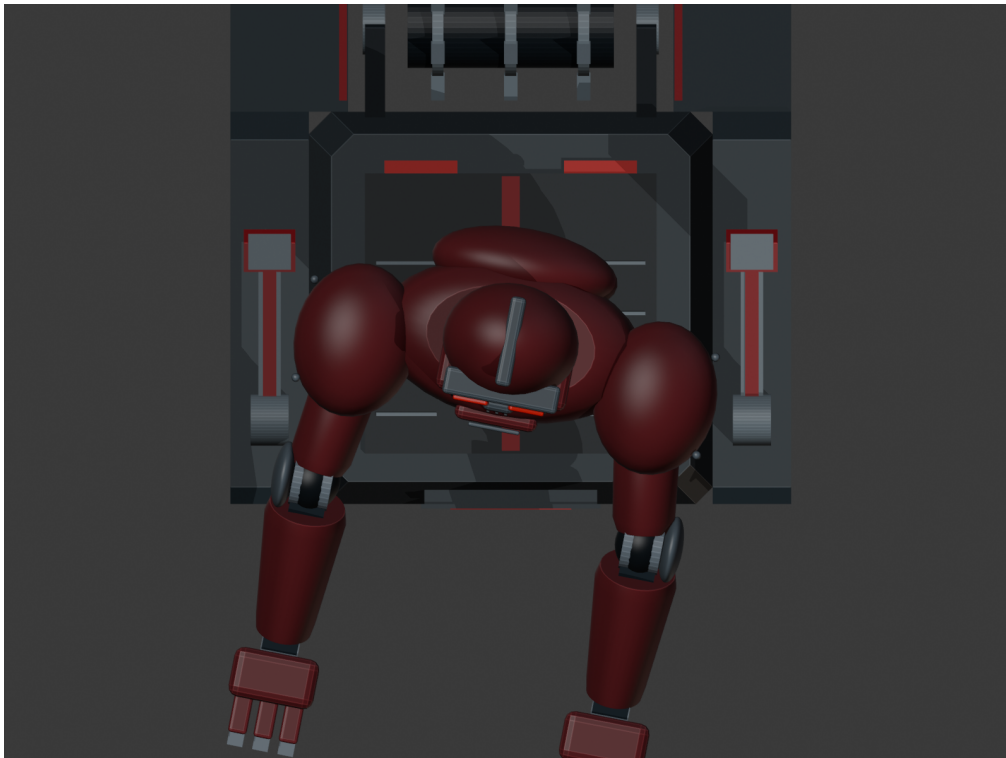


Figure 4. Top CAD view used as part of dimensional verification.

9. Engineering Mass Budget

The CAD meshes do not contain verified physical mass metadata, so I use an explicit engineering subsystem allocation rather than claiming a false simulated mass.

Subsystem	Planned mass
Chassis and lower armor	1.10 kg
Drive motors and gearboxes	0.75 kg
Wheels, hubs and shafts	0.30 kg
Battery and power system	0.60 kg
RC electronics, ESC and wiring	0.25 kg
Guarded active mechanism + drive / mount	0.70 kg
Upper structure, arms and outer shells	0.50 kg
Bearings, brackets and fasteners	0.25 kg
Contingency allowance	0.20 kg
TOTAL	4.650 kg

The remaining design allowance is 0.350 kg below 5.000 kg. The completed physical robot must be weighed on a calibrated scale before operational use.

10. Bill of Materials and Cost

The Bill of Materials records component name, quantity, unit cost, supplier, supplier stock code, direct purchase URL and total line cost. The current priced component subtotal is R8,039.15. The guarded active-mechanism motor / controller remains intentionally marked TBD until its electrical and mass compatibility is finalized.

Core item	Cost status
6 mm HDPE chassis / armor stock	Priced
Drive motors and wheels	Priced
2.4 GHz transmitter + receiver	Priced
3S LiPo battery	Priced
Dual brushed drive ESC	Priced
Main isolation, E-Stop, relay, fuse and distribution	Priced
High-current wiring and fasteners	Priced
Guarded active-mechanism motor / controller	TBD
Current priced component subtotal	R8,039.15

BOM file: [SentinelT_Bill_of_Materials.xlsx](#)

11. Buildability and Serviceability

SentinelT is designed as a modular build. Removable panels provide access to the battery, receiver, ESC, power distribution, main isolation and E-Stop control hardware. The selected direct-drive wheels mount to 6 mm motor shafts, reducing separate drivetrain parts.

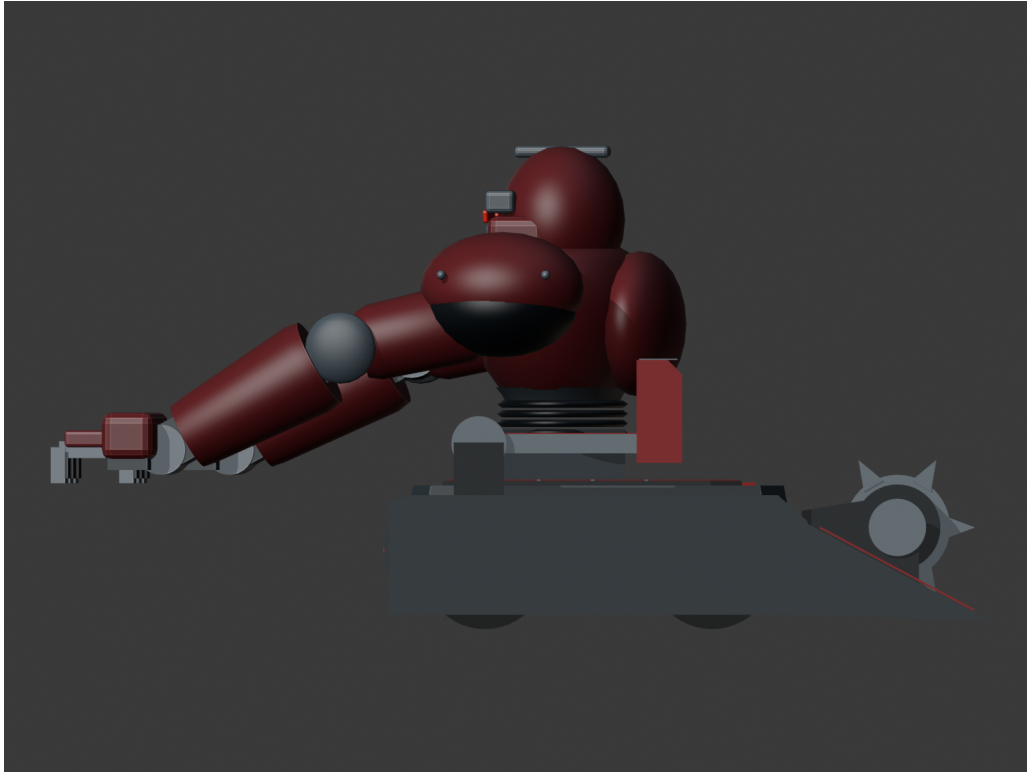


Figure 5. Side CAD view showing the compact chassis envelope.

12. Current Development Status

The digital robot design, external CAD geometry, dimensional simulation, control framework, safety architecture and procurement plan are documented. The next CAD revision adds simplified internal component models, wiring paths, a separate 2.4 GHz transmitter model, a cutaway view and an exploded internal-layout view.

Physical claims remain separated from digital results. Final physical mass, current draw, thermal behaviour, drivetrain response and E-Stop interruption performance will be measured after fabrication and bench testing.

13. Project Ownership and Responsibilities

SentinelT is independently designed and developed by Thato Glen Assegaai. My responsibilities include system concept, Blender CAD modelling, mechanical packaging, Python-based verification tools, 2.4 GHz RC control framework, safety architecture, component research, Bill of Materials development, technical documentation and repository organisation.

Third-party commercial components are identified by manufacturer and supplier names in the Bill of Materials. Their inclusion does not imply ownership of those products; the SentinelT system design, integration plan, CAD work, control framework and documentation are presented as my project work.